

Dual-CellNucDETR: Context-Aware Object-Level Cell Analysis

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Abstract. Cell detection and classification are fundamental tasks in computational pathology. Recent DETR-based methods formulate nucleus analysis as direct object detection, simplifying the prediction pipeline while achieving competitive performance. Although larger fields of view improve cell classification by providing broader tissue context, scaling a single DETR detector requires substantially more object queries, increasing computational cost.

We propose Dual-CellNucDETR, a dual-stage Deformable DETR architecture that decouples nucleus localisation from contextual classification. A first stage detects nuclei on local image patches, while a second stage performs contextual classification over all detected nuclei using the complete image, enabling larger fields of view without scaling the detector. Experiments on PUMA and Internal Lung histopathology datasets show that the proposed approach achieves competitive or superior classification performance while substantially reducing memory consumption compared with a single-stage DETR operating on full-resolution images.

Keywords: Computational pathology · Object detection · Deformable DETR · Cell nuclei analysis · Histopathology

1 Introduction

Accurate cell detection and classification in histopathological whole-slide images (WSIs) are fundamental tasks in computational pathology. Reliable identification of individual cells enables quantitative characterisation of the tumour microenvironment, biomarker discovery, patient stratification, and clinically relevant decision support. As digital pathology becomes increasingly adopted in clinical and research practice, there is a growing demand for automated methods capable of analysing large tissue images accurately and efficiently.

Over the last decade, instance segmentation has become the dominant paradigm for automated cell analysis. Architectures such as HoVer-Net [3], HoVer-NeXt [1], and CellViT [4] jointly segment individual nuclei and assign a cell type, achieving

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state-of-the-art performance across public benchmarks. More recently, object detection has emerged as an alternative formulation for cell analysis. DETR-based architectures formulate nucleus analysis as a direct set prediction problem, predicting each nucleus as an individual object rather than as part of a dense segmentation map. This formulation considerably simplifies the prediction pipeline while maintaining competitive detection and classification performance, making it an attractive alternative for large-scale computational pathology applications.

Despite these advances, most current cell analysis methods are trained and evaluated on relatively small image patches, typically around 256×256 pixels. While this field of view is generally sufficient for accurate nucleus localisation, cell classification often depends not only on nuclear morphology but also on broader contextual cues, including neighbouring cells, tissue organisation, and the local microenvironment. Consequently, limiting the available field of view may hinder the classification of morphologically similar or ambiguous cell types.

A straightforward way to exploit broader tissue context is to increase the input field of view. However, larger inputs also increase the computational cost. In segmentation-based methods, this mainly results in higher memory consumption and longer inference times. DETR-based detectors face an additional challenge: larger fields of view contain substantially more nuclei, requiring many more object queries to maintain detection performance. Consequently, scaling DETR-based detectors rapidly becomes computationally expensive. This raises a fundamental question: can larger tissue context be exploited without scaling the detector itself?

To address this limitation, we propose Dual-CellNucDETR, a dual-stage DETR-based architecture that decouples nucleus localisation from contextual cell classification. Rather than scaling a single detector to process a large field of view, the proposed architecture performs localisation and contextual classification at their most appropriate spatial scales. A first stage detects nuclei on local 256×256 image patches, while a second DETR decoder performs contextual classification over the complete 1024×1024 image, enabling global contextual reasoning without requiring a large increase in object queries.

Our main contributions are as follows:

- **Impact of field of view on DETR-based cell classification.** We show that larger fields of view improve classification performance but require substantially more object queries, increasing computational cost.
- **Dual-CellNucDETR architecture.** We propose a dual-stage DETR architecture that decouples nucleus localisation from contextual classification, enabling larger fields of view without scaling a single detector.
- **Comprehensive experimental evaluation.** We evaluate Dual-CellNucDETR against state-of-the-art segmentation- and DETR-based methods on two 1024×1024 histopathology datasets, demonstrating a competitive trade-off between classification performance and computational efficiency.

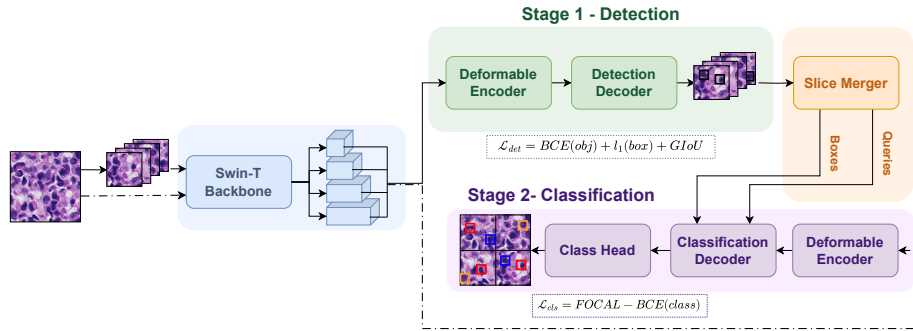


Fig. 1: Dual-CellNucDETR architecture

2 Related Work

Automated cell analysis in histopathology has traditionally been formulated as an instance segmentation problem, where nuclei are first delineated and subsequently assigned a cell type. HoVer-Net [3] established this paradigm by jointly predicting nuclei segmentation masks, geometric offset maps, and cell classes, recovering individual nuclei through watershed-based post-processing. HoVer-NeXt [1] retained the same segmentation-first formulation while simplifying the architecture and post-processing pipeline to improve computational efficiency. More recently, CellViT [4] replaced the convolutional encoder with a Vision Transformer backbone while maintaining the same segmentation-first paradigm.

An alternative paradigm formulates nucleus analysis as an object detection task. DETR [2] casts object detection as a direct set prediction problem using transformer decoders and learned object queries. Deformable DETR [8] extends this framework with sparse multi-scale attention, substantially improving convergence speed and detection efficiency. Building upon this architecture, CellNucDETR [6] adapts the framework to jointly detect and classify cell nuclei.

Despite their architectural differences, both segmentation- and DETR-based methods are typically trained on small image patches, around 256×256 pixels. While this field of view is generally sufficient for accurate nucleus localisation, it limits the tissue context available for cell classification. Increasing the input size is a natural solution, but substantially increases the computational cost of segmentation models and, in DETR-based detectors, requires many more decoder object queries.

3 Methodology

3.1 Architecture

CellNucDETR: CellNucDETR formulates cell nuclei analysis as a direct object detection problem by adapting Deformable DETR to histopathology images. A hierarchical Swin Transformer [5] backbone extracts multi-scale features,

which are processed by a Deformable DETR encoder. A Deformable DETR decoder then operates on a fixed set of learned object queries, predicting the location and class of each nucleus through deformable cross-attention.

Dual-CellNucDETR: While CellNucDETR jointly performs localisation and classification within a single Deformable DETR decoder, scaling this formulation to larger fields of view requires substantially more object queries, increasing computational cost and memory consumption. At the same time, cell classification benefits from the broader tissue context available in larger images.

To address this limitation, we propose Dual-CellNucDETR (Figure 1), which decouples nucleus localisation from contextual classification into two specialised Deformable DETR stages. The first stage detects nuclei on local 256×256 image tiles, whereas the second stage classifies the detected nuclei using the complete 1024×1024 image. Both stages share the same backbone and Deformable DETR architecture as CellNucDETR but operate at different spatial scales.

Detection stage. Each 1024×1024 image is divided into overlapping 256×256 tiles, which are processed independently by the detection branch of CellNucDETR using a decoder with $Q = 200$ learned queries. The detector predicts nucleus bounding boxes and binary objectness scores.

Proposal merging. Detections from all tiles are reprojected to the original image and merged into a single set of nucleus proposals. Low-confidence detections and spurious large boxes are removed using score and aspect-ratio thresholds, followed by non-maximum suppression. The remaining proposals are ranked by objectness, and the top- K detections together with their query embeddings are forwarded to the second stage. We use $K = 1500$ for the 1024×1024 experiments.

Classification stage. The second stage receives the complete 1024×1024 image together with the top- K proposals generated by the first stage. Instead of randomly initialised queries, the decoder is initialised from the Stage-1 query embeddings, allowing each query to correspond to a detected nucleus. The decoder then predicts the final cell class from $C + 1$ classes using the broader tissue context available in the full image. The additional background class allows false-positive proposals from the first stage to be discarded.

3.2 Losses

Dual-CellNucDETR is trained with separate detection and classification losses. The final objective is their weighted combination.

Detection loss. It supervises Stage 1 on 256×256 tiles using Hungarian matching. The loss combines L_1 , binary cross-entropy (BCE), and Generalized Intersection over Union (GIoU) for box regression and objectness prediction.

$$\mathcal{L}_{\text{det}} = \lambda_{L1} L1 + \lambda_{\text{GIoU}} (1 - \text{GIoU}) + \lambda_{\text{obj}} \text{BCE} \quad (1)$$

Classification Loss: It supervises Stage 2 on the full 1024×1024 image. The loss is computed over the valid proposals $\mathcal{V}$ and the C cell classes using a multi-label BCE formulation. To address class imbalance, the focal term $\mathcal{F}$

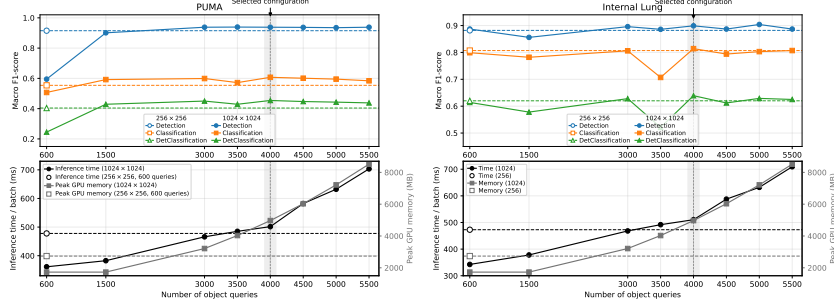


Fig. 2: Performance of CellNucDETR with different decoder-query configurations on PUMA (left) and Internal Lung (right). Detection, classification, and Det-Classification macro F₁-scores, GPU memory consumption, and inference time.

down-weights easy examples, while the positive class weights w_c^+ compensate for class imbalance.

$$\mathcal{L}_{\text{cls}} = \frac{1}{|\mathcal{V}|C} \sum_{\mathcal{V}} \sum_c \mathcal{F} \cdot \text{BCE}_{w_c^+} \quad (2)$$

4 Experimental Setup

4.1 Datasets

PUMA: This public benchmark [7] contains 1024×1024 H&E-stained melanoma WSIs extracted at $40\times$ magnification from 100 primary and 100 metastatic cases. The original ten cell categories were grouped into five PanNuke-compatible classes: Neoplastic, Inflammatory, Connective, Necrosis, and Epithelial.

Internal Lung: This dataset contains 139 H&E-stained lung images (1024×1024 , $40\times$). Each nucleus is annotated with a bounding box and labelled as Tumour or Non-tumour.

Both datasets were split into training, validation, and test subsets (80%, 5%, 15%) and standardised using the training-set mean and standard deviation.

4.2 Evaluation Protocol

Baselines: We compare Dual-CellNucDETR against the segmentation-based methods HoVer-Net, HoVer-NeXt, and CellViT, as well as the DETR-based CellNucDETR. All models were initialised from the publicly available PanNuke-pretrained checkpoints and fine-tuned on each target dataset following their official implementations. HoVer-NeXt, CellViT, and CellNucDETR were additionally trained using 1024×1024 images to evaluate larger fields of view.

Evaluation metrics: Predicted and ground-truth nuclei are matched using the Hungarian algorithm on nucleus centroids with Euclidean distance as the cost.

Table 1: Macro-averaged detection, classification, and DetClassification metrics.

Method	Detection			Classification			DetClassification		
	Precision	Recall	F ₁ -score	Precision	Recall	F ₁ -score	Precision	Recall	F ₁ -score
Dataset: PUMA									
HoVer-Net (256)	0.84	0.93	0.89	0.49	0.51	0.50	0.33	0.37	0.35
HoVer-NeXt (256)	0.91	0.90	0.90	0.57	0.56	0.55	0.41	0.40	0.39
HoVer-NeXt (1024)	0.88	0.91	0.90	0.70	0.49	0.49	0.56	0.37	0.36
CellViT (256)	0.90	0.84	0.87	0.55	0.53	0.54	0.40	0.37	0.38
CellViT (1024)	0.94	0.91	0.92	0.60	0.52	0.52	0.45	0.39	0.39
CellNucDETR (256)	0.89	0.93	0.91	0.60	0.53	0.55	0.43	0.40	0.40
CellNucDETR (1024)	0.92	0.95	0.93	0.62	0.59	0.60	0.46	0.45	0.45
Dual-CellNucDETR	0.93	0.95	0.94	0.63	0.57	0.59	0.47	0.44	0.44
Dataset: Internal Lung									
HoVer-Net (256)	0.92	0.82	0.86	0.89	0.81	0.83	0.75	0.60	0.65
HoVer-NeXt (256)	0.90	0.84	0.87	0.86	0.82	0.83	0.69	0.61	0.65
HoVer-NeXt (1024)	0.95	0.85	0.90	0.84	0.79	0.81	0.70	0.60	0.64
CellViT (256)	0.94	0.76	0.84	0.82	0.76	0.78	0.66	0.51	0.57
CellViT (1024)	0.95	0.82	0.88	0.91	0.81	0.84	0.80	0.60	0.66
CellNucDETR (256)	0.93	0.83	0.88	0.85	0.78	0.80	0.70	0.57	0.62
CellNucDETR (1024)	0.93	0.86	0.90	0.83	0.80	0.81	0.68	0.60	0.63
Dual-CellNucDETR	0.90	0.89	0.89	0.89	0.82	0.84	0.72	0.64	0.67

A match within a radius of $r = 12$ pixels, as used in HoVer-Net [3], is considered a true positive. Detection performance is reported using Precision, Recall, and F₁-score, whereas classification is evaluated only on correctly detected nuclei and reported using macro-averaged Precision, Recall, and F₁-score. Following the HoVer-Net evaluation protocol, we additionally report the macro-averaged DetClassification F₁-score, which jointly evaluates detection and classification by accounting for both classification errors and unmatched detections.

Efficiency evaluation: Inference time and peak GPU memory were measured on a single NVIDIA A40 GPU (48 GB VRAM). For a fair comparison, models operating on 256×256 and 1024×1024 images were evaluated with batch sizes of 64 and 4, respectively.

5 Experiments and results

5.1 Does increasing the field of view improve cell classification?

We investigate the effect of increasing the field of view on CellNucDETR by training different decoder-query configurations on 1024×1024 images and comparing them with the original 256×256 model (Figure 2).

On the PUMA dataset (left), the standard 600-query configuration is insufficient to match the performance of the original 256×256 model, indicating that larger fields of view require additional decoder capacity. Detection rapidly recovers as the number of queries increases, whereas classification and DetClassification continue improving up to approximately 4000 queries, after which gains

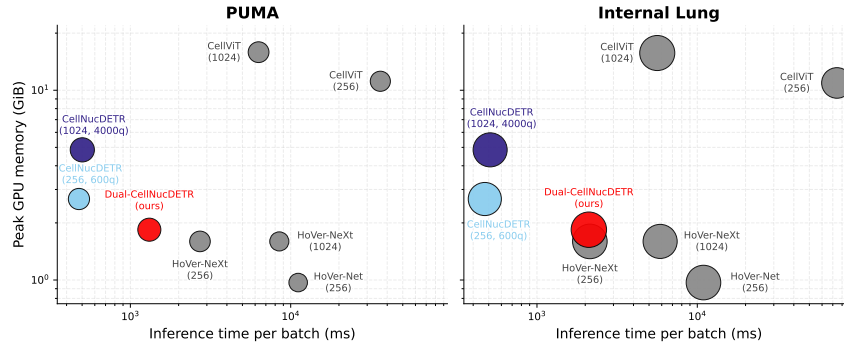


Fig. 3: Performance-efficiency comparison on PUMA (left) and Internal Lung (right). Bubble area denotes the macro DetClassification F_1 -score; axes show inference time and peak GPU memory.

become marginal. A similar behaviour is observed on the Internal Lung dataset (right), although the effect is less pronounced. Detection remains largely stable across query configurations, whereas classification and DetClassification peak at around 4000 queries, with little benefit thereafter. The lower plots show that inference time and GPU memory consumption increase steadily with the number of decoder queries. Overall, larger fields of view improve classification but require substantially higher computational cost.

5.2 Dual-CellNucDETR

Table 1 compares Dual-CellNucDETR with state-of-the-art methods. HoVer-NeXt, CellViT, and CellNucDETR were additionally trained on 1024×1024 images to assess larger fields of view. On the PUMA dataset, increasing the field of view does not consistently improve the segmentation-based baselines. In contrast, CellNucDETR achieves the best classification and DetClassification performance, while Dual-CellNucDETR obtains the highest detection F_1 -score and remains competitive on both classification metrics. On the Internal Lung dataset, the effect of larger fields of view is more architecture dependent. While HoVer-NeXt improves detection at the expense of classification and CellViT benefits in both tasks, Dual-CellNucDETR achieves the highest DetClassification F_1 -score while matching the best classification performance.

5.3 Performance-efficiency trade-off

Figure 3 compares the evaluated methods in terms of DetClassification F_1 -score, inference time, and GPU memory consumption. Dual-CellNucDETR achieves a favourable performance-efficiency trade-off on both datasets. Compared with CellNucDETR operating on 1024×1024 images, it substantially reduces GPU

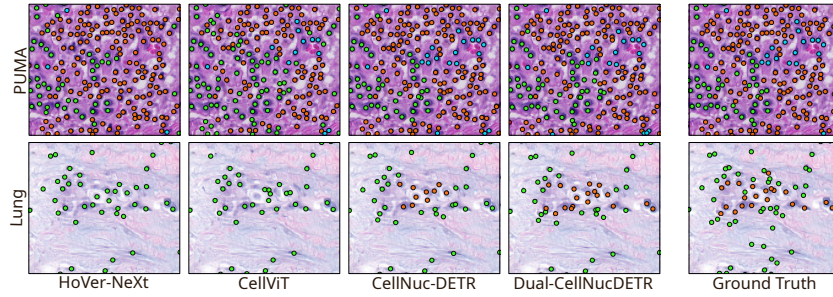


Fig. 4: Qualitative comparison. Top: PUMA (green: Neoplastic, orange: Inflammatory, cyan: Connective, red: Necrosis, white: Epithelial). Bottom: Internal Lung (green: Non-tumour, orange: Tumour).

memory consumption while maintaining comparable DetClassification performance. This reduction comes at the cost of longer inference time due to the sequential execution of the detection and contextual classification stages.

5.4 Qualitative analysis

Figure 4 presents representative predictions on the PUMA and Internal Lung datasets. On the PUMA dataset, HoVer-NeXt and CellViT recover the overall cellular structure but exhibit more frequent class confusions in mixed regions. CellNucDETR produces more consistent predictions, while Dual-CellNucDETR further reduces these confusions. On the Internal Lung dataset, HoVer-NeXt and CellViT correctly localise most nuclei but tend to over-predict the dominant non-tumour class. CellNucDETR improves tumour identification, whereas Dual-CellNucDETR provides the closest agreement with the ground truth, consistent with its superior DetClassification performance.

5.5 Discussion

The results support the central hypothesis of this work: broader tissue context improves cell classification, but scaling a single DETR detector to larger fields of view is computationally inefficient. By decoupling localisation from contextual classification, Dual-CellNucDETR exploits global contextual information without requiring thousands of object queries. A class-wise analysis (not shown) suggests that the gains on PUMA mainly arise from inflammatory, connective, and necrosis cells, indicating that tissue context is particularly beneficial for morphologically challenging cell types. The main limitation of the proposed approach is the increased inference time introduced by the two-stage design.

6 Conclusions

We investigated how larger fields of view can be effectively exploited for object-based cell analysis. Although additional tissue context improves cell classifica-

tion, scaling a single DETR detector to larger images requires substantially more object queries, increasing computational cost.

To address this limitation, we proposed Dual-CellNucDETR, which decouples nucleus localisation from contextual classification. Experiments on two datasets show that this strategy exploits broader tissue context while substantially reducing GPU memory consumption. These findings suggest that decoupling localisation and contextual classification is a promising strategy for context-aware cell analysis.

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